Zomato Data Analysis: SQL Output Report

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1. About This Report

This report shows the SQL outputs for the Zomato Data Analysis project. First the data was checked and cleaned (Part A). Then it was analysed to find business insights (Part B). Each section gives the task, what was found, and what it means in simple words.

2. Part A: Data Exploration

2.1 Duplicate Check on RestaurantID

Task: Check if any RestaurantID appears more than once.

Method: Group the rows by RestaurantID and keep only the IDs that appear more than one time.

Output: The query returned **0 rows**.

RestaurantID	id_count
(no rows returned)	-

Finding: The table has 9,551 rows, and no RestaurantID is repeated.

Conclusion: Every restaurant is listed only once, so no rows needed to be deleted.

2.2 Column Removal

Task: Find and remove columns that are not needed, so the table is smaller and cleaner.

Method: Each column was compared with the project tasks. A column that no task uses, or that repeats another column, was marked for removal. The remaining columns were then checked with SQL counts.

Columns removed and the reason:

Column removed	Reason
Address	No task uses the full street address. City and Locality are enough for the analysis.

Column removed	Reason
LocalityVerbose	It repeats Locality with the city name added. The City column already stores the city, so it adds no new information.
Switch_to_order_menu	No task uses it.
Is_delivering_now	No task uses it, and it only shows the status at one moment in time.

Evidence for LocalityVerbose (count check output):

total_rows	locality_count	locality_verbose_count	unique_localities	unique_verbose_localities
9551	9551	9551	1208	1265

What the output shows: Both columns are filled in every row, so neither has missing values. LocalityVerbose has 1,265 unique values and Locality has 1,208. The extra 57 values appear only because the city name is attached to the locality, so the same locality in two cities is split into two values. The City column already holds that information, so LocalityVerbose can be removed.

Columns kept: RestaurantID and RestaurantName (identity), CountryCode (needed to join the country table), City and Locality (location analysis), Cuisines, Currency, Has_Table_booking, Has_Online_delivery, Price_range, Votes, Average_Cost_for_two, and Rating (all used in the analysis tasks).

Conclusion: Removing four columns made the table simpler without losing any information needed for the tasks.

2.3 Correcting Misspelled City Names

Task: Find and correct misspelled city names.

Method: Three checks were run on the City column.

Check	What it looks for	Output
TRIM check	Extra spaces before or after a name	No rows returned
SOUNDEX check	Names that sound alike	2 pairs returned
Special character check (REGEXP)	Names with non-English or special characters	4 names returned

SOUNDEX output:

sound_code	variants	n
P230	Pasay City, Pasig City	2
P524	Panchkula, Pensacola	2

Special character output:

City	Meaning
Brasília	Correct spelling, accent letter
São Paulo	Correct spelling, accent letter
İstanbul	Correct spelling, special Turkish letter
Cedar Rapids/Iowa City	Correct name, it is one listed area written with a slash

(In the exported output file, the accent letters appear as a question mark because of character encoding. The actual city names in the table are correct.)

Finding:

- Pasay City and Pasig City are two different cities in the Philippines.
- Panchkula is in India and Pensacola is in the USA, so they are also different places.
- The four special character names are valid spellings.

Conclusion: All checks showed that the city names were already consistent. No UPDATE was needed.

2.4 Rolling Count Analysis

Task: Use a window function to calculate a running total of restaurants.

Method: Restaurants were counted for each city, and then a running total was added in alphabetical order of city using `SUM(COUNT(*)) OVER (ORDER BY City)`.

Sample of the output (141 cities in total):

City	Restaurants	Rolling count
Abu Dhabi	20	20
Agra	20	40
Ahmedabad	21	61
Albany	20	81
Allahabad	20	101
New Delhi	5473	7963
Noida	1080	9043
Winchester Bay	1	9550
Yorkton	1	9551

Check: The last rolling count is **9,551**. This is the same as the total number of rows, so the query is correct.

Finding: The data covers 141 cities. New Delhi alone has 5,473 restaurants, which is more than half of the dataset.

Conclusion: The running total grows city by city and ends exactly at the total number of restaurants.

3. Part B: Data Analysis

3.1 Country Distribution

Task: Find the percentage of restaurants in each country.

Method: After joining the country table to get Country_Name, restaurants were counted per country and divided by the total of 9,551.

Country	Total restaurants	Percentage
India	8,652	90.59%
United States	434	4.54%

Country	Total restaurants	Percentage
United Kingdom	80	0.84%
Brazil	60	0.63%
United Arab Emirates	60	0.63%
South Africa	60	0.63%
New Zealand	40	0.42%
Turkey	34	0.36%
Australia	24	0.25%
Philippines	22	0.23%
Indonesia	21	0.22%
Singapore	20	0.21%
Qatar	20	0.21%
Sri Lanka	20	0.21%
Canada	4	0.04%

Finding: The 15 countries add up to 9,551 restaurants. India has about 91% of the data, and the United States is second with about 4.5%. The percentages total 100.01% only because each value is rounded to two decimals.

Conclusion: The dataset is heavily focused on India, so most results will reflect Indian restaurants.

3.2 Popular Cities: Top Three by Number of Restaurants

Task: Find the cities with the most listed restaurants.

Method: Restaurants were counted per city (from the rolling count output) and sorted from highest to lowest.

Rank	City	Restaurants
1	New Delhi	5,473
2	Gurgaon	1,118
3	Noida	1,080

Finding: The top three cities are all in the Delhi region, and together they hold 7,671 restaurants, about 80% of the data. New Delhi is far ahead of the others.

Conclusion: The Delhi region is the strongest area for Zomato in this dataset.

3.3 Restaurant Recommendations

Task: Find the best moderately priced restaurants using cost, rating, votes, and facilities.

Output (top 10):

Restaurant	City	Cost for two	Rating	Votes	Table booking	Online delivery
Caterspoint	Gurgaon	500	4.9	223	No	Yes
CakeBee	Coimbatore	350	4.9	200	No	Yes
Truffles	Bangalore	800	4.7	9,667	No	Yes
Tapri Central	Jaipur	750	4.7	1,469	No	Yes
Coffee Cup	Secunderabad	800	4.6	1,408	Yes	No
India Restaurant	Kolkata	800	4.6	1,219	Yes	Yes
Spezia Bistro	New Delhi	900	4.6	1,071	Yes	Yes
Twigly	Gurgaon	500	4.6	395	No	Yes
Joey's Pizza	Mumbai	800	4.5	2,662	No	Yes
Burgrill	Chandigarh	500	4.5	249	No	Yes

Finding:

- Every restaurant in the list has a rating of 4.5 or higher, a cost for two between 350 and 900, and at least 200 votes.
- Truffles in Bangalore has the most votes (9,667) with a high rating of 4.7, so its rating is the most trusted.
- Only India Restaurant, Spezia Bistro, and Coffee Cup offer table booking. Most of the list offers online delivery.

Best restaurant in Kolkata: India Restaurant, with a rating of 4.6, 1,219 votes, a cost for two of 800, and both table booking and online delivery.

Conclusion: Good moderately priced restaurants combine a high rating with a healthy number of votes, and online delivery is common among them.

4. Overall Conclusion

- The data is clean: there are no duplicate RestaurantIDs and no misspelled city names.

- Four unneeded columns were removed (Address, LocalityVerbose, Switch_to_order_menu, Is_delivering_now).
- The rolling count ends at 9,551, which matches the total rows.
- India holds about 91% of the restaurants, and the Delhi region (New Delhi, Gurgaon, Noida) holds about 80%.
- The best moderately priced restaurants have high ratings, many votes, and mostly offer online delivery.

5. Assumptions

- The rolling count was calculated city by city in alphabetical order, because the project does not say how to group it.
- The special character results were checked by hand to confirm they are valid names.
- "Moderately priced" means a cost for two between 350 and 900, as shown in the recommendation output.